

Sustainability points

For draft versions of letters or photos, use 'draft' mode, which saves ink and toner, and often increases printing speeds as well. You may find that you are satisfied with less than the highest-resolution mode even for final copies.

You can print work in progress in smaller size, or print several pages on the same page in order to save ink and paper.

Print a multi-page document as smaller as 'thumbnails' with four or more pages on a sheet. This will help you to get a pre-final view of printed pages.

Preview before printing. The print preview feature of most software programs lets you see what a printout will look like so that you can avoid a false start. Print selectively. Instead of printing webpages, bookmark them or 'Save As' to a default folder. If you do print them, use mono option, not colour.

If you leave your printer on standby mode without using it much, you will still run up a high electricity bill. Inkjet printers use power even when they are switched off but plugged into the mains. So printers must be disconnected when not in use.

Some models allow you to ePrint to the printer from anywhere, by e-mailing your document to it or using a mobile app. Some Wi-Fi-equipped

printers have mobile-device applets available for download that allow Apple iOS and Android-based mobile devices to print via the Wi-Fi network.

Print quality. For 8x10 colour photos, the simplest 'best print' mode was used when a setting was offered. Some printer drivers enabled 'fixing' features (colour and contrast correction) as defaults that could be changed, but in the test these advanced settings in the printer driver weren't tried to tweak photos. The manufacturer's best recommended glossy photo paper was used with each printer.

For photos, the panelists were instructed to give four scores: one for clarity (including mechanical defects), one for contrast and brightness, one for colour and an overall score. They were given a set of three references for each score at 3, 6 and 9, and additional scored references with other specified defects in each category were available as well. They were also instructed to give an overall score for each photo.

Dell 2155cn performed best in printing quality.

Timing. Printing times were recorded for the Dr Grauert Letter—a standardised letter used to judge the print performance. It was printed five times and the time in seconds was recorded for the first page and after five pages. The same document was then used with one word changed from black to red. The time for five pages was recorded. Printing time was recorded for the colour spreadsheet. One print was timed in the normal mode on plain paper. Similarly, the DTP document was timed using the appropriate mode for premium matte paper.

Scanning. Scans were made and timed—using default settings for each target—for a magazine page, an 8x10 colour photo and a 3D object. The scanned images were viewed on a colour-calibrated 61cm (24-

inch) Eizo Color Edge LCD monitor. Three expert viewers made consensus judgments on the scans, using a high-quality scan as the reference. HP Envy 100 e-All-in-One D410a performed best and HP Photosmart e-AIO B010a worst in this test. Scanning speed was recorded for a text document, a magazine page and a colour photo.

Copying. The copying capability was tested using the magazine page. The computer was not used and all settings were made from the printer. The magazine page was copied using best settings for colour copy and normal for B&W copy. Times were recorded for both. HP Officejet 7500A wide-format e-AIO E910a and Dell 2155cn brands performed best, while the performance of Dell 1355cnw was lowest.

Faxing. The degree of problems, if any, seen in a faxed business letter sent and received through a reference fax device were rated by a lab expert. All the printers performed well in this test.

Ink consumption. The cartridge used for each test and its weight were recorded. The number of pages printed and the cost-per-page calculations were recorded. Note that the cost of appropriate paper should be taken into account when calculating the full cost per print. The cost of a sheet of quality photo paper can be twice that of the ink needed to print the photo. For snapshot printers that are tested with a combo package of ink and paper, the cost of the paper is included.

Noise. None of the printers is silent; most inkjets make typical 'swish-ing' printer noise as they print and 'clunking' sounds as they feed the paper, and can produce additional cartridge maintenance noises before and after printing. The judgments take into account the disturbing qualities of these transient sounds, as well as the loudness and length of the cooling-fan sounds on laser printers. HP Envy 100 e-All-in-One D410a and Brother DCP-J715W were the least noisy.

Power consumption. Power consumption in watts was measured



Dell 2155cn multifunction colour laser printer